Engineering economics:

4.1 Consider two alternative water resource projects, A and B. Project A will cost $2,533,000 and will return $1,000,000 at the end of 5 years and $4,000,000 at the end of 10 years. Project B will cost $4,000,000 and will return $2,000,000 at the end of 5 and 15 years, and another $3,000,000 at the end of 10 years. Project A has a life of 10 years, and B has a life of 15 years. Assuming an interest rate of 0.1 (10%) per year:

(a) What is the present value of each project?
(b) What is each project's annual net benefit?
(c) Would the preferred project differ if the interest rates were 0.05?
(d) Assuming that each of these projects would be replaced with a similar project having the same time stream of costs and returns, show that by extending each series of projects to a common terminal year (e.g., 30 years), the annual net benefits of each series of projects will be same as found in part (b).

(a) Present Value of A = $\{-2.533 + 1.0(1.10)^{-5} + 4.0(1.10)^{-10}\}10^6$
$= -369{,}900$

Present Value of B = $\{-4.0 + 2.0[(1.10)^{-5} + (1.10)^{-15}] + 3.0(1.10)^{-10}\}10^6$
$= -1{,}122{,}700$

(b) Net Annual Benefit of A = $(-370{,}000)(\dfrac{0.1(1.1)^{10}}{(1.1)^{10}-1}) = -60{,}200$

Net Annual Benefit of B = $(-1{,}123{,}000)(\dfrac{0.1(1.1)^{15}}{(1.1)^{15}-1}) = -147{,}600$

(c) Present Value of A = $\{-2.533 + 1.0(1.05)^{-5} + 4.0(1.05)^{-10}\}10^6 = 706{,}200$

Present Value of B = $\{-4.0 + 2.0[(1.05)^{-5} + (1.05)^{-15}] + 3.0(1.05)^{-10}\}10^6$
$= 370{,}800$

Net Annual Benefit of A = $(706{,}200)(\dfrac{0.05(1.05)^{10}}{(1.05)^{10}-1}) = 91{,}500$

Net Annual Benefit of B = $(370{,}800)(\dfrac{0.05(1.05)^{15}}{(1.05)^{15}-1}) = 35{,}700$

Preferred project is A at 5% interest. Neither is preferred at 10% interest, but A is preferable to B.

(d) Payment of 369, 900 at beginning of years 1, 11 and 21 for A has a present value of –567,500, which in turn has a 30-year net annual benefit of –60,200 (same as (b) above). Payments of 1,122,700 at the beginning of years 1 and 16 have a present value of –

1,391,500, which in turn has a 30-year net annual benefit of –147,600 (same as (b) above).

4.9 Assume water can be allocated to three users. The allocation, x_j, to each use j provides the following returns: $R(x_1) = (12\, x_1 - x_1^2)$, $R(x_2) = (8\, x_2 - x_2^2)$ and $R(x_3) = (18\, x_3 - 3\, x_3^2)$. Assume that the objective is to maximize the total return, $F(\boldsymbol{X})$, from all three allocations and that the sum of all allocations cannot exceed 10.

a) How much would each use like to have?

b) Show that at the maximum total return solution the marginal values, $\partial(R(x_j))/\partial x_j$, are each equal to the shadow price or Lagrange multiplier (dual variable) λ associated with the constraint on the amount of water available.

c) Finally, without resolving a Lagrange multiplier problem, what would the solution be if 15 units of water were available to allocate to the three users and what would be the value of the Lagrange multiplier?

a) If each x_j is unrestricted, optimal allocation = 6, 4 and 3 for j =1, 2, and 3 respectively.

b) If $\sum_{j=1}^{3} x_j \le 10$, optimal allocations are 33/7, 19/7, 18/7 respectively and

$\lambda = {}^{18}\!/_{7} = \frac{dF}{dx_j}\Big|_{x_j optimal}$ where F is the original objective function.

c) If $\sum_{j=1}^{3} x_j \le 15$, then the solution is 6, 4, and 3 with $\lambda = 0$ since $6 + 4 + 3 \le 15$.

This should be obvious from the first part of this exercise so no computational work needed. Excess resource is available.

4.15 Consider a three-season reservoir operation problem. The inflows are 10, 50 and 20 in seasons 1, 2 and 3 respectively. Find the operating policy that minimizes the sum of total squared deviations from a constant storage target of 20 and a constant release target of 25 in each of the three seasons. Develop a discrete dynamic programming model that considers only 4 discrete storage values: 0, 10, 20 and 30. Assume the releases cannot be less than 10 or greater than 40. Show how the model's recursive equations change depending on whether the decisions are the releases or the final storage volumes. Verify the optimal operating policy is the same regardless of whether the decision variables are the releases or the final storage volumes in each period. Which model do you think is easier to solve? How would each model change if more importance were given to the desired releases than to the desired storage volumes?

The general recursion equation assuming final storage volume decisions is:

$$f_t^n(s_t) = \text{minimum}\{(20-s_t)^2 + (25-(s_t+i_t-s_{t+1}))^2 + f_{t+1}^{n-1}(s_{t+1})\} \quad \text{for } 0 \le s_t \le K$$

$$0 \le s_{t+1} \le K$$
$$s_{t+1} \le s_t + i_t$$
$$10 \le + s_t + i_t - s_{t+1} \le 40$$

The general recursion equation assuming release decisions is:

$$f_t^n(s_t) = \text{minimum}\{(20-s_t)^2 + (25-r_t)^2 + f_{t+1}^{n-1}(s_{t+1})\} \quad \text{for } 0 \le s_t \le K$$

$$10 \le r_t \le s_t + i_t \le 40$$
$$r_t \le s_t + i_t - K$$
$$s_t + i_t - r_t = s_{t+1}$$

Note, the dynamic programming network remains the same in both cases. The nodes are storage values, the links releases. The solution will be the same whether the decision variable is a release or final storage, since knowing one implies knowing the other.

Multiplying the second term of the objective function by a weight greater than one will give more importance to meeting the release targets. Alternatively, the exponent of the second term of objective could be increased if kept an even integer number.

Optimal final storage volumes and releases given initial storage volume S:

S	R_1	S_2	R_2	S_3	R_3	S_1
0	10	0	30	20	10	10
10	10,20	10,0	30	30	10	20
20	20	10	40	30	20	20
30	30	10	--	--	30	20

More detail on the solution at each iteration, using a backward-moving algorithm and final storage (S_{t+1}) as a decision variable is:

$n = 1$		
S_3	$f_3^1(S_3)$	S_1^*
0	425	0
10	125	0,10
20	25	10,20
30	125	20,30

$n = 2$		
S_2	$f_2^2(S_2)$	S_3^*
0	450	20
10	250	20
20	350	30
30	-	-

$n = 3$		
S_1	$f_1^3(S_1)$	S_2^*
0	1075	0
10	575	0,10
20	275	10
30	375	10

$n = 4$		
S_3	$f_3^4(S_3)$	S_1^*
0	1200	10
10	600	20
20	300	20
30	400	30

$n = 5$		
S_2	$f_2^5(S_2)$	S_3^*
0	725	20
10	525	30
20	625	30
30	-	-

$n = 6$		
S_1	$f_1^6(S)$	S_2^*
0	1350	0
10	850	0,10
20	550	10
30	650	10

$n = 7$		
S_3	$f_3^7(S_3)$	S_1^*
0	1475	10
10	875	20
20	575	20
30	675	20

Note: Steady-state solution is reached after first iteration

4.18 The following table provides estimates for the recent values of the costs of additional wastewater treatment plant capacity needed at the end of each 5-year period for the next 20 years. Find the capacity expansion schedule that minimizes the present values of the total future costs. If there is more than one least cost solution, indicate which one you think is better, and why?

		DISCOUNTED COST OF ADDITIONAL CAPACITY: UNITS OF ADDITIONAL CAPACITY					Total Required Capacity at End of Period
Period	*Years*	*2*	*4*	*6*	*8*	*10*	
1	1-5	12	15	18	23	26	2
2	6-10	8	11	13	15		6
3	11-15	6	8				8
4	16-20	4					10

The cost in each period t must be paid at the beginning of the period. What was the discount factor used to convert the costs at the beginning of each period t to present value costs shown above? In other words how would a cost at the beginning of period t be discounted to the beginning of period 1, given an annual interest rate of r? (Only the algebraic expression of the discount factor is asked, not the numerical value of r.)

Let S_t = existing capacity at the beginning of period t
D_t = required capacity at the end of period t
X_t = capacity addition in period t
$f_t(S_t)$ = minimum total discounted cost of future capacity expansion through period 4 given S_t at beginning of period t.

Recursive equations:

$$f_5(10) = 0$$

$$f_t(S_t) = \underset{\substack{X_t \\ 0 \le X_t \le D_t - S_t}}{\text{minimum}}\{Cost_t(X_t) + f_{t+1}(S_t + X_t)\} \qquad D_{t-1} \le S_t \le 10$$

Solutions:

Stage T	State S_t	$Cost_t(X_t) + f_{t+1}(S_t + X_t)$ X_t: 0	2	4	6	8	10	$f_t(S_t)$	X_t
4	8	-	4					4	2
	10	0						0	0
3	6	-	10	8				8	4
	8	4	6					4	0
	10	0						0	0
2	2	-	-	19	17	15		15	8
	4	-	16	15	13			13	6
	6	8	12	11				8	0
	8	4	8					4	0
	10	0						0	0
1	0	-	27	28	26	27	26	26	6,10

Hence two solutions costing 26 are

	X_1	X_2	X_3	X_4
Solution *A*	6	0	4	0
Solution *B*	10	0	0	0

Solution *A* permits a greater opportunity to adjust to changing (especially decreasing) future demands for capacity and costs of capacity expansion. The discount factor for each of the four periods t is at = $(1 + r)^{-5(t-1)}$, where r is the annual interest rate expressed as a fraction.

This problem can be easily solved by drawing the appropriate network of links (feasible decisions) from each feasible state (possible existing capacity).